REDKROSS RESEARCH FOUNDATION

LINUX SECURITY BASICS

Linux Workstation Hardening & Security Assessment

SUBMITTED BY	Vibhas Deepak Pawar
ORGANIZATION	Redkross Research Foundation
ENVIRONMENT	Ubuntu Linux virtual machine
PLATFORM	Oracle VirtualBox on Windows
TASK TYPE	Practical Linux security and system hardening

INTERNSHIP PRACTICAL REPORT

Prepared to document the security checks performed in an Ubuntu virtual lab. Observations are limited to commands and results reported during the practical.

1. Introduction

Linux is widely used in servers, workstations, cloud computing, and cybersecurity environments. Although Linux includes built-in security mechanisms, proper configuration and regular maintenance are essential to reduce exposure to unauthorized access, malicious software, and known vulnerabilities.

This task focused on fundamental Linux security practices in an Ubuntu virtual machine. Activities included package maintenance, firewall configuration, file-permission inspection, service review, system-log inspection, network-socket inspection, and verification of AppArmor.

2. Objectives

- Understand the importance of Linux workstation security.
- Refresh package information and check available software upgrades.
- Configure and verify the UFW firewall.
- Inspect permissions on important system account files.
- Review administrative privileges and the principle of least privilege.
- Check SSH availability and review system services.
- Inspect system logs and listening network sockets.
- Verify automatic-update settings and the unattended-upgrades service.
- Check AppArmor status and enforcement.
- Document observations and identify follow-up actions.

3. Tools and Environment

Component	Purpose
Windows host	Host operating system
Oracle VirtualBox	Runs the Ubuntu virtual machine
Ubuntu Linux	Practical security lab environment
APT	Package management and software updates
UFW	Host-based firewall configuration
AppArmor	Mandatory access control for supported applications
Linux Terminal tools	Service, log, permission, and network checks

4. Practical Implementation

4.1 System Package Updates

Objective: Refresh package information and install eligible software updates.

```
sudo apt update<br/>sudo apt upgrade -y
```

The package index was refreshed and an upgrade was attempted. During final verification, two packages remained listed as upgradable: **orca** and **xserver-common**. A normal upgrade did not install them. They were left unchanged rather than forcing a broader upgrade that might modify or remove additional packages.

Observation: Package management commands were used, but two package upgrades remained pending. Regular updates help address known vulnerabilities; the pending packages should be investigated and updated when

appropriate.

4.2 Firewall Configuration Using UFW

Objective: Enable the firewall and verify its configuration.

```
sudo ufw status<br/>sudo ufw enable<br/>sudo ufw status verbose
```

The firewall was enabled and its status was checked. The reported final configuration showed **Status: active**, the default incoming policy set to **deny**, and logging enabled at low level.

Result: The firewall was active and incoming connections were denied by default. Required services should be permitted only through deliberate firewall rules.

4.3 Automatic Updates

Objective: Check periodic-update configuration and the unattended-upgrades service.

```
cat /etc/apt/apt.conf.d/20auto-upgrades<br/>sudo service unattended-upgrades status
```

The two relevant settings were reported as set to **1**:

```
APT::Periodic::Update-Package-Lists "1";<br/>APT::Periodic::Unattended-Upgrade "1";
```

The unattended-upgrades service was reported as active and running.

Result: The periodic-update settings were enabled and the service was active. Periodic checks should still confirm that updates are completing successfully.

4.4 User Account and File Permissions

Objective: Inspect permissions on important Linux account files.

```
ls -l /etc/passwd /etc/shadow
```

The permissions of both files were inspected. **/etc/passwd** stores general account information, while **/etc/shadow** stores password-hash information and related password data. The displayed permissions were reported to match the expected patterns. No manual permission changes were made.

Security relevance: Appropriate ownership and permissions help prevent unauthorized changes to account data and unauthorized access to password hashes.

4.5 Least Privilege

Objective: Review the current user's group memberships.

```
groups
```

The output included the **sudo** group. Membership in this group allows administrative commands to be run with elevated privileges when authorized.

Recommendation: Use elevated privileges only when necessary and avoid granting unnecessary administrative access.

4.6 SSH Service Review

Objective: Determine whether the SSH service was available.

```
sudo service ssh status
```

The system reported that **ssh.service** could not be found. SSH was not available under the checked service name, and no SSH installation or remote-access configuration was performed.

Security relevance: SSH should be enabled only when remote administration is required. If introduced later, use secure authentication, restrict access, and configure appropriate firewall rules.

4.7 System Service Review

Objective: Review service status to identify services that may affect system function and security.

```
service --status-all 2>&1 | tee ~/service-review.txt
```

The service review included entries such as AppArmor, UFW, unattended-upgrades, cron, D-Bus, CUPS, OpenVPN, and Chrony. The output was saved to **~/service-review.txt**. Services were reviewed, and no service was disabled solely because it appeared in the list.

Security relevance: Disabling unnecessary services can reduce the attack surface, but each service should be assessed for purpose and dependencies before any change.

4.8 System Log Inspection

Objective: Inspect recent system messages.

```
sudo tail -n 20 /var/log/syslog
```

Recent system-log entries were displayed and inspected. Logs can contain service events, warnings, errors, and other operational information.

Limitation: This was a basic log review, not a complete security-log audit.

4.9 Listening Network Sockets

Objective: Inspect listening TCP and UDP sockets.

```
sudo ss -tulpn
```

The command was executed to inspect listening sockets and associated processes where available. Individual ports were not recorded in the final notes.

Follow-up: Review the output and investigate any unexpected listening ports to confirm that each exposed service is required.

4.10 AppArmor Verification

Objective: Verify the status of AppArmor.

```
sudo aa-status
```

The output confirmed that the AppArmor module was loaded and that profiles were in enforce mode. Profiles in enforce mode apply their configured restrictions to the applications and processes covered by those profiles.

Result: AppArmor was loaded, and enforcement was confirmed for profiles reported in enforce mode.

5. Summary of Results

Security area	Observation	Status
Package management	Two packages remained upgradable: orca and xserver-common.	Follow-up required
UFW firewall	Active; incoming traffic denied by default.	Verified
Automatic-update settings	Both relevant periodic settings were set to 1.	Verified
Unattended upgrades	Service active and running.	Verified
Account-file permissions	Expected patterns reported; no changes made.	Checked
Administrative privileges	User membership in sudo group identified.	Reviewed
SSH	Service not found under the checked name.	Reviewed
System services	Service list reviewed and saved.	Completed
System logs	Recent log entries inspected.	Completed
Listening sockets	Inspection command executed; individual ports not recorded.	Check performed
AppArmor	Module loaded; profiles in enforce mode.	Verified

6. Security Recommendations

- **Maintain regular updates:** Check for package updates and investigate packages that remain unupgraded.
- **Maintain firewall protection:** Keep UFW enabled and allow only required network traffic.
- **Verify automatic updates:** Check update configuration and review whether updates are completing successfully.
- **Protect account files:** Preserve correct ownership and permissions for /etc/passwd and /etc/shadow.
- **Apply least privilege:** Use administrative privileges only when necessary and review privileged accounts.
- **Review services:** Disable services only after confirming they are unnecessary and that dependencies will not be affected.
- **Secure remote access:** Do not enable SSH without a requirement; if required, configure authentication and firewall rules securely.
- **Monitor logs:** Review relevant logs for recurring errors, unusual events, and security warnings.
- **Inspect network exposure:** Investigate listening ports and confirm that each service is needed.
- **Maintain AppArmor:** Keep AppArmor enabled and review enforcement status periodically.

7. Learning Outcomes

- Used Linux terminal commands for basic system administration.
- Refreshed package indexes and attempted software upgrades with APT.
- Enabled and verified a Linux firewall.
- Inspected important account-file permissions.
- Reviewed administrative group membership and system services.
- Inspected system logs and listening network sockets.
- Checked automatic-update settings and service status.
- Verified AppArmor loading and profile enforcement.
- Practised documenting observations and avoiding unverified system changes.

8. Conclusion

This task provided practical exposure to essential Linux workstation-hardening techniques in an Ubuntu virtual machine. The work covered package management, firewall configuration, automatic-update settings, account-file permissions, service review, log inspection, listening network sockets, and AppArmor.

The firewall was confirmed active with incoming connections denied by default. The unattended-upgrades service was active, the relevant periodic-update settings were enabled, and AppArmor was loaded with profiles in enforce mode. Two packages—orca and xserver-common—remained unupgraded during the final package check and require follow-up investigation.

The practical strengthened understanding of Linux administration and the importance of verifying security settings, documenting limitations, and maintaining a cautious approach to system hardening.

Appendix A. Evidence Checklist

Attach genuine screenshots captured from the Ubuntu virtual machine. The screenshots should show the actual command and its output; do not recreate or edit command results.

- **Evidence 1 — Firewall:** Capture the output of `sudo ufw status verbose`, showing the active status and default incoming policy.
- **Evidence 2 — AppArmor:** Capture the relevant output from `sudo aa-status`, showing that the module is loaded and profiles are in enforce mode.
- **Evidence 3 — Automatic updates:** Capture the output of `cat /etc/apt/apt.conf.d/20auto-upgrades`, showing both periodic settings set to 1.

Appendix B. Pending Package Note

At the time of the final check, APT listed **orca** and **xserver-common** as upgradable, but a normal upgrade reported zero packages upgraded and two not upgraded. These packages were not force-upgraded as part of this report. Their dependency or upgrade requirements should be reviewed before making further changes.